

Complex Power Series

Complex Analysis

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Definition

If (f_n) is a sequence of functions defined on a subset Ω of $\mathbb{C}$ with values in $\mathbb{C}$, the sequence of **partial sums** (s_n) of the infinite series $\sum f_n$ is defined for z in Ω by

$$s_1(z) := f_1(z)$$

$$s_2(z) := s_1(z) + f_2(z)$$

$$\vdots$$

$$s_{n+1}(z) := s_n(z) + f_n(z)$$

$$\vdots$$

In case the sequence (s_n) of functions converges on Ω to a function f , we say that the infinite series of functions $\sum f_n$ **converges** to f on Ω . We will often write

$$\sum f_n \quad \text{or} \quad \sum_{n=1}^{\infty} f_n$$

to denote either the series or the limit function, when it exists.

Definition

If the series $\sum |f_n(z)|$ converges for each $z \in \Omega$, we say that $\sum f_n$ is **absolutely convergent** on Ω . If the sequence (s_n) of partial sums is uniformly convergent on Ω to f , we say that $\sum f_n$ is **uniformly convergent** on Ω , or that it **converges to f uniformly on Ω** .

Definition

A series of complex functions $\sum f_n$ is said to be a **power series around** $z = z_0$ if the function f_n has the form

$$f_n = a_n(z - z_0)^n,$$

where a_n and z_0 belong to $\mathbb{C}$ and where $n = 0, 1, 2, \dots$

Cauchy-Hadamard Formula

Definition

Let $\{a_n\}$ be a sequence. If the sequence $(|a_n|^{1/n})$ is bounded, we set $\rho := \limsup(|a_n|^{1/n})$; if this sequence is not bounded we set $\rho = \infty$. We define the **radius of convergence** of $\sum a_n(z - z_0)^n$ to be given by

$$R := \begin{cases} 0 & \text{if } \rho = \infty, \\ 1/\rho & \text{if } 0 < \rho < \infty, \\ \infty & \text{if } \rho = 0. \end{cases}$$

This is known as the **Cauchy-Hadamard formula**.

If R is the radius of convergence of a power series around z_0 , then the **disc of convergence** is the set $D_R(z_0)$.

Cauchy-Hadamard Theorem

Theorem (Cauchy-Hadamard Theorem)

If R is the radius of convergence of the power series $\sum a_n(z - z_0)^n$, then the series is absolutely convergent if $|z - z_0| < R$ and is divergent if $|z - z_0| > R$.

Remark

It should be noted that the Cauchy-Hadamard Theorem makes no statement as to whether the power series converges when $|z| = R$. Indeed, anything can happen, as the examples

$$\sum x^n, \quad \sum \frac{1}{n} x^n, \quad \sum \frac{1}{n^2} x^n, \quad x \in \mathbb{R}$$

show. Since $\lim(n^{1/n}) = 1$, each of these power series has radius of convergence 1. The first power series converges at neither of the points $x = -1$ and $x = 1$; the second series converges at $x = -1$ but diverges at $x = 1$; and the third power series converges at both $x = -1$ and $x = 1$.

Theorem

Let R be the radius of convergence of $\sum a_n(z - z_0)^n$ and let $\mathcal{D}$ be any closed disk contained in $D_R(z_0)$. Then the power series converges uniformly on $\mathcal{D}$.

Corollary

Let R be the radius of convergence of $\sum a_n(z - z_0)^n$. Then for each $0 < r < R$, the series converges uniformly on $\overline{B}_r(z_0)$.

Example

Consider the series

$$\sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}, \quad \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}.$$

The radius of convergence of each series is infinite ($R = \infty$), that is, each series converges (absolutely and uniformly) on $\mathbb{C}$.